

EXPERIMENT 9

Implement Dielectric free space boundary condition in Electric Field and find the relative permittivity of the different medium.

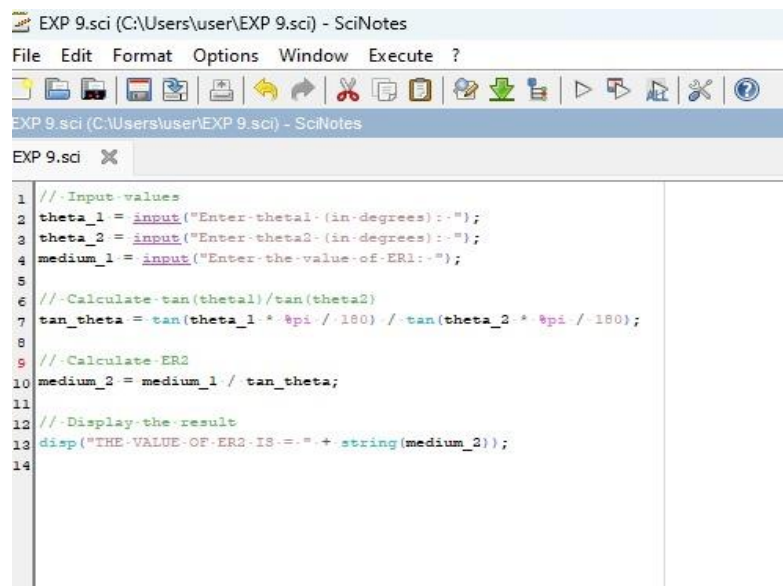
Aim:

To write program for Dielectric – free space boundary condition in Electric field using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:



```
1 //Input-values
2 theta_1 = input("Enter-theta1- (in-degrees) :-");
3 theta_2 = input("Enter-theta2- (in-degrees) :-");
4 medium_1 = input("Enter-the-value-of-ER1:-");
5
6 //Calculate-tan(theta1)/tan(theta2)
7 tan_theta = tan(theta_1.*%pi./180) ./ tan(theta_2.*%pi./180);
8
9 //Calculate-ER2
10 medium_2 = medium_1 / tan_theta;
11
12 //Display-the-result
13 disp("THE-VALUE-OF-ER2-IS:-" + string(medium_2));
14
```

Output:

Enter thetal (in degrees): 30

Enter theta2 (in degrees): 70

Enter the value of ER1: 8

"THE VALUE OF ER2 IS = 38.070164"

Theoretical calculations:

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Output:

Enter the value of theta 1 = 30°

Enter the value of theta 2 = 70°

Enter the value of ER1 = 8

Practical output:

"The value of ER2 IS = 38.070164"

Theoretical calculations:

For dielectric boundary condition:

$$\frac{\epsilon_1}{\epsilon_2} = \frac{\tan \theta_2}{\tan \theta_1}$$

Therefore,

$$\epsilon_1 = \epsilon_2 \frac{\tan \theta_2}{\tan \theta_1}$$
$$\epsilon_1 = 8 \times \frac{\tan 70^\circ}{\tan 30^\circ}$$
$$\tan 70^\circ = 2.7475, \quad \tan 30^\circ = 0.57735$$
$$\epsilon_1 = 8 \times \frac{2.7475}{0.57735}$$
$$\epsilon_2 = 8 \times 4.759$$
$$\epsilon_2 = 38.0707$$

Theoretical value $\epsilon_2 = 38.0707$

Case 1: In this case, the dielectric medium has a relative permittivity (ϵ_r) of 2, meaning the

Dielectric reduces the electric field by half compared to the free space. This is typical for

Materials like mica or certain types of plastics.

- Free Space Electric Field (E_1): 100 V/m
- Dielectric Electric Field (E_2): 50 V/m

Case 2: Here, the dielectric medium has a higher relative permittivity ($\epsilon_r=2.5$) compared to

Case 1. This indicates a stronger polarization effect in the dielectric, as seen in materials like

Glass or water.

- Parameters:

○ $E_{n1}=200$ V/m (Electric field in free space).

○ $E_{n2}=80$ V/m (Electric field in the dielectric medium).

Case 3: The dielectric medium here also has a relative permittivity of 2. The behavior of the

Material suggests it could be another low-polarization dielectric material, with a consistent

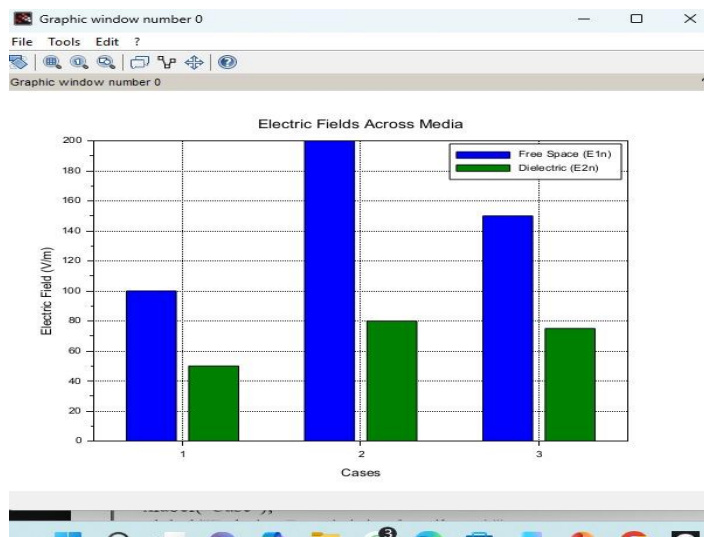
Reduction of the electric field by half.

- Free Space Electric Field (E_{1n}): 150 V/m
- Dielectric Electric Field (E_{2n}): 75 V/m

Program:

```
EXP 9.sci (C:\Users\user\EXP 9.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 9.sci (C:\Users\user\EXP 9.sci) - SciNotes
EXP 9.sci x
3 clear;
4 clf;
5
6 // Constant
7 epsilon0 = 8.854e-12; // Permittivity of free space (F/m)
8
9 // Case Parameters
10 E1n = [100 200 150]; // Electric field in free space (V/m)
11 E2n = [50 80 75]; // Electric field in dielectric (V/m)
12
13 // Compute Relative Permittivity
14 epsilon_r = E1n ./ E2n;
15
16 // Display Results
17 disp("Relative Permittivity for Each Case:");
18 disp(epsilon_r);
19
20 // Case numbers
21 n = [1 2 3];
22
23 // -----
24 // Bar Graph
25 // -----
26 hcf(0);
27 bar(n, [E1n' E2n']);
28
29 xlabel("Cases");
30 ylabel("Electric Field (V/m)");
31 title("Electric Fields Across Media");
32 legend(["Free Space (E1n)", "Dielectric (E2n)"]);
33 grid();
34
35 // -----
36 // Line Graph
37 // -----
38 hcf(1);
39 plot(n, epsilon_r, "-o");
40
41 xlabel("Cases");
42 ylabel("Relative Permittivity");
43 title("Relative Permittivity for Each Case");
44 grid();
45
46 // Display values
47 disp("Case-1: ER = " + string(epsilon_r(1)));
48 disp("Case-2: ER = " + string(epsilon_r(2)));
49 disp("Case-3: ER = " + string(epsilon_r(3)));
50
```

Output:



Theoretical calculations:

1. Case 1:

$$\varepsilon_r = \frac{E_{1n}}{E_{2n}} = \frac{100}{50} = 2$$

2. Case 2:

$$\varepsilon_r = \frac{E_{1n}}{E_{2n}} = \frac{200}{80} = 2.5$$

3. Case 3:

$$\varepsilon_r = \frac{E_{1n}}{E_{2n}} = \frac{150}{75} = 2$$

Result:

Thus the program was verified for Dielectric – free space boundary condition in Electric field using SCILAB was successfully.